

PROBLEM STATEMENT NO 4- SUPERSTORE SALES DATASET

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Course: MBA (Finance & Business Analytics)

Mapped CO4: Construct visualization using appropriate charts and graphs

Data Visualization and Dashboard Analysis

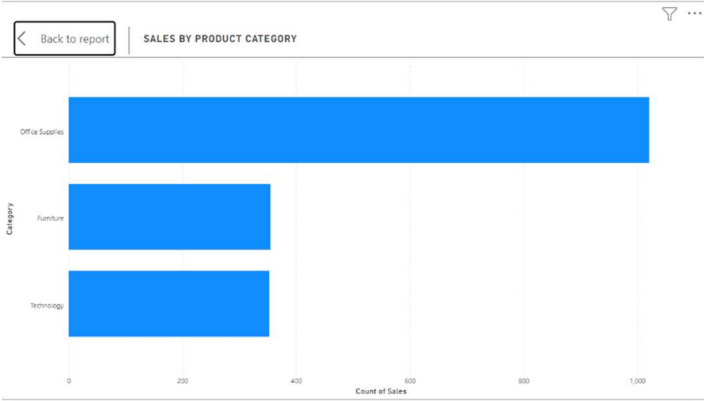
The cleaned Superstore dataset was used to create a dashboard for understanding sales performance from different perspectives. The charts focus on product categories, monthly sales, sub-categories, regions, states, and customer segments. Each visual presents a different part of the dataset and helps make the results easier to understand.

1.Sales by Product Category:

Objective:

This chart compares the sales count across the three main product categories: Office Supplies, Furniture, and Technology. It helps identify which category has the highest level of sales activity.

Chart Type: Horizontal Bar Chart



Observation:

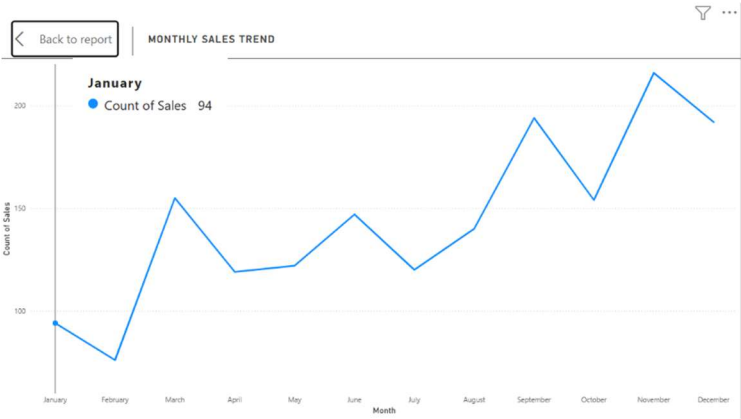
Office Supplies shows the highest sales count in the dashboard, while Furniture and Technology have comparatively lower values. This gives a quick view of the difference in sales activity between the major categories.

2.Monthly Sales Trend:

Objective:

The monthly sales chart is used to observe how sales change from January to September. It helps identify months with higher or lower sales activity.

Chart Type: Line Chart



Observation:

The sales pattern changes from month to month rather than remaining constant. Sales are lower during February and

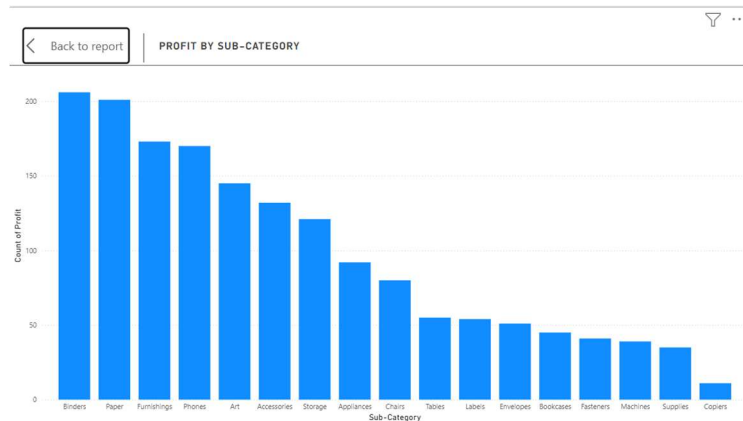
increase noticeably in March. Another increase can be seen towards September, which records the highest value shown in the chart.

3.Profit by Sub-Category:

Objective:

This chart compares profit across different product sub-categories. It helps identify the sub-categories that contribute more to the overall profit value.

Chart Type: Column Chart



Observation:

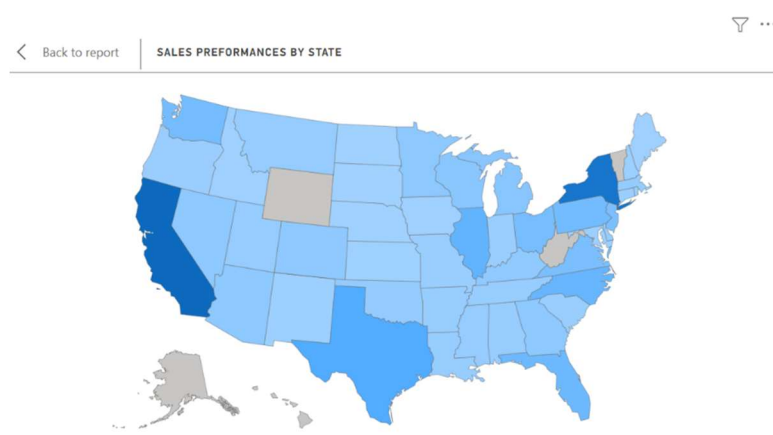
Binders and Paper appear at the higher end of the chart, followed by Furnishings and Phones. Tables and Chairs show comparatively lower values. This comparison can help management understand which sub-categories require closer attention.

4. Sales Performance by State:

Objective:

The Shape Map is used to understand how sales are distributed across different states. It provides a simple geographical view of sales performance and makes it easier to compare states with higher and lower sales.

Chart Type: Shape Map



Observation:

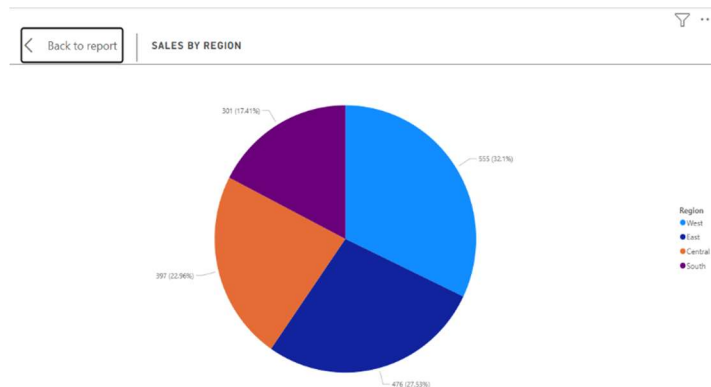
The map shows differences in sales across the states. States with higher sales are represented by stronger shades, while states with lower sales appear in lighter shades. This makes it easier to identify areas with better sales performance and states that may need further attention.

5. Sales by Region:

Objective:

The pie chart is used to show how sales are divided among the different regions in the Superstore dataset. It helps compare the contribution of each region to the overall sales.

Chart Type: Pie Chart



Observation:

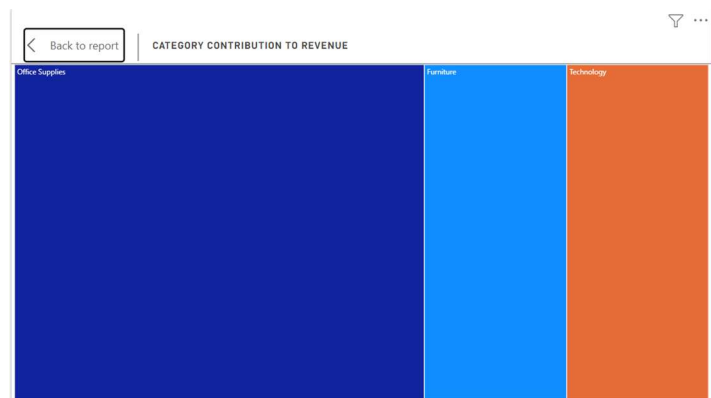
The chart shows that the West region has the largest share of sales, while the South region has the smallest share among the four regions. The East and Central regions contribute the remaining portions of total sales.

6. Category Contribution to Revenue:

Objective:

The treemap shows how each product category contributes to the overall revenue. The size of each section represents its relative contribution.

Chart Type: Treemap



Observation:

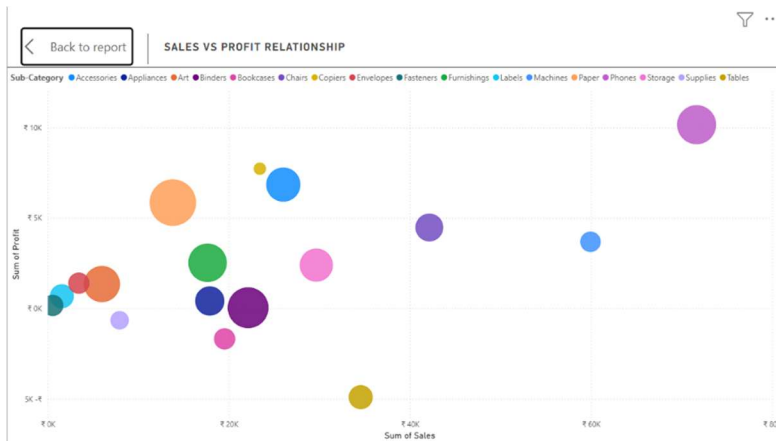
Office Supplies occupies the largest area in the treemap, indicating its higher contribution compared with Furniture and Technology. Furniture and Technology form smaller portions of the total revenue shown.

7. Sales vs. Profit Relationship:

Objective:

This visual examines the relationship between sales and profit for different sub-categories. It can also help identify records where higher sales do not necessarily result in higher profit.

Chart Type: Scatter Plot



Observation:

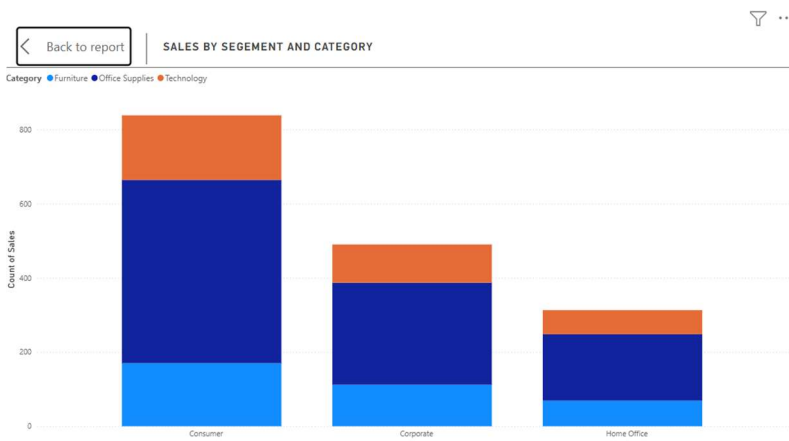
Most of the plotted values are concentrated within a relatively small profit range. Some points are positioned away from the main group, indicating differences between individual sub-categories. This type of chart makes unusual sales and profit combinations easier to notice.

8.Sales by Segment and Category:

Objective:

This chart compares sales across the Consumer, Corporate, and Home Office segments while also showing the contribution of Furniture, Office Supplies, and Technology within each segment.

Chart Type: Stacked Column Chart



Observation:

The Consumer segment has the highest sales count among the three segments. Corporate follows, while Home Office has a smaller total. Office Supplies forms a significant portion of sales across the segments, with Furniture and Technology making up the remaining portions.